

Blind rigid-motion CBCT — the problem

one measurement, two unknowns: the geometry cannot be fitted without the image, and the image cannot be solved without the geometry

$$y = A_{P(\theta)} x + n, \quad P(\theta) = P_{nom} T(\theta)$$

the cone-beam forward model. A_P is the projector for the V per-view matrices P ; $T(\theta)$ is the rigid motion.

(1)

$$y \in \mathbb{R}^{V \times n_v \times n_u}, \quad x \in \mathbb{R}^N, \quad \theta \in \mathbb{R}^{V \times 6}$$

$V=360$ views, $n_v \times n_u = 500 \times 700$, $N=256^3$:
126 M measurements, 16.8 M + 2 160 unknowns.

(2)

$$(\hat{x}, \hat{\theta}) = \arg \min_{x, \theta} \frac{1}{2} \|A_{P(\theta)} x - y\|^2 + \lambda \text{TV}(x) + \mathcal{R}_\psi(x)$$

the joint (blind) problem. It is BILINEAR in (x, θ) , hence non-convex: many minima explain the same y .

(3)

$$\Theta(x) \equiv \arg \min_{\theta} \frac{1}{2} \|A_{P(\theta)} x - y\|^2$$

GEOMETRY given the image. Measured here: from the cold FDK every estimator config plateaus at $\approx 2.0^\circ$; from a clean reference the same budget reaches $0.09\text{--}0.27^\circ$.

(4)

$$X(\theta) \equiv \arg \min_x \frac{1}{2} \|A_{P(\theta)} x - y\|^2 + \lambda \text{TV}(x)$$

IMAGE given the geometry. Measured here: the same loop reaches 40.45 dB with the true θ from step 0, 38.16 dB blind.

(5)

$$\hat{\theta} = \Theta(X(\hat{\theta})), \quad \hat{x} = X(\Theta(\hat{x}))$$

THE CIRCULARITY, stated exactly: each unknown is defined through the other, so neither map can be evaluated on its own.

(6)

$$A_{P(\theta \circ g^{-1})}(g \cdot x) = A_{P(\theta)} x \quad \forall g \in SE(3)$$

an EXACT gauge: a global rigid pose is unobservable, so the solution is an ORBIT $\{(g \cdot \hat{x}, \hat{\theta} \circ g^{-1})\}$, not a point.

(7)

$$\text{depth along the beam: } \partial y / \partial t_{\parallel} \approx 0$$

and beyond the gauge, translation ALONG THE BEAM is nearly unobservable — 92% of a converged estimator's residual lives there.

(8)